

Defining elliptic curves on $\mathbb{C}$ from lattices

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17 July 2025

1 Lattices and elliptic functions

Definition 1.1. Let $w_1, w_2 \in \mathbb{C}$ such that they are not linearly independent, a *lattice* $L = [w_1, w_2]$ is an additive subgroup generated by complex number w_1 and w_2 , i.e.

$$[w_1, w_2] := w_1\mathbb{Z} + w_2\mathbb{Z} = \{w_1a + w_2b \in \mathbb{C} \mid a, b \in \mathbb{Z}\}$$

Remark 1.2. Let L be a lattice:

- i) since $(\mathbb{C}, +)$ is an abelian group, it has all normal subgroups. Then we can consider $\mathbb{C}/L$ a subgroup;
- ii) since $\mathbb{C}$ has the euclidean topology and L has the discrete one, we can also see $\mathbb{C}/L$ as a topological space, the complex torus ¹.

Definition 1.3. Let $a, w_1, w_2 \in \mathbb{C}$ and L a lattice. A *fundamental parallelogram* for L with vertex a is the following set

$$\mathcal{F}_a := \left\{ a + t_1w_1 + t_2w_2 \mid \begin{array}{l} t_1 \in [0, 1) \\ t_2 \in [0, 1) \end{array} \right\}$$

Definition 1.4. Let L be a lattice, an *elliptic function* f is a complex function f such that:

- i) f is meromorphic on $\mathbb{C}$;
- ii) f is periodic with respect to L , i.e. $f(z + w) = f(z)$ for any $w \in L$.

Remark 1.5. Given an elliptic function $f : \mathbb{C} \rightarrow \mathbb{C}$, the periodicity implies that f induces a function $f^\#$ such that:

$$\begin{array}{ccc} f^\# : \mathbb{C}/L & \rightarrow & \mathbb{C}/L \\ [z] & \mapsto & [f(z)] \end{array}$$

Then we can see the elliptic functions as functions on $\mathbb{C}/L$.

¹It can be proved that the complex torus is a compact topological manifold.

2 Eisenstein series and the Weierstrass $\wp$ -function

Lemma 2.1. For any lattice L and integer $k > 2$ the series $\sum_{w \in L \setminus \{0\}} \frac{1}{w^k}$ converges absolutely.

The previous lemma allows us to give the following definition

Definition 2.2. Let L be a lattice and $k \in \mathbb{N}$ such that $k > 2$. The *weight- k Eisenstein series* for L is the sum

$$G_k(L) := \sum_{w \in L^*} \frac{1}{w^k}$$

where $L^* := L \setminus \{0\}$

Remark 2.3. Let $w \in L$, then also $-w \in L$. Now let us suppose that k is odd, then we have that the terms $\frac{1}{w^k}$ and $\frac{1}{(-w)^k}$ in the sum cancel, then $G_k(L) = 0$. Thus the only interesting Eisenstein series are those of even weight.

Definition 2.4. Given a lattice L , the *Weierstrass $\wp$ -function* of L is defined by

$$\wp(z, L) := \frac{1}{z^2} + \sum_{w \in L^*} \left(\frac{1}{(z-w)^2} - \frac{1}{w^2} \right)$$

The following result allows us to give the previous definition

Theorem 2.5. For any lattice L , the series $\wp(z, L)$ converges and is holomorphic in $\mathbb{C} \setminus L$.

Since $\wp$ converges absolutely we can compute the derivative by differentiating term by term and we have that

$$\wp'(z) = -2 \sum_{w \in L} \frac{1}{(z-w)^3}$$

It is clear also that $\wp$ and $\wp'$ have respectively poles of order 2 and 3 in each point of L .

Remark 2.6. Since $|L| = |\mathbb{Z} \times \mathbb{Z}| = \aleph_0$ we can choose an order, induced by the bijectivity between L and $\mathbb{N}$, such that for both $\wp$ and its derivative we can do the sum. Moreover, the sum does not depend by the choice of the bijectivity (and then by the order on L), indeed, since the series converges absolutely, we have the commutativity property also for an infinity number of addend. Hence, both of the functions are well defined.

One can also prove that both $\wp$ and $\wp'$ are periodic with respect to L , then we have

Proposition 2.7. $\wp$ and $\wp'$ are elliptic functions for each lattice L .

Moreover L is a discrete set, then exists $r \in \mathbb{R}_{>0}$ such that $\wp$ is holomorphic in $B_r(0) \setminus \{0\}$. Then it follows that $\wp$ admits a Laurent series expansion in centered in 0, indeed we have the following result:

Lemma 2.8. Let L be a lattice, the Laurent series expansion for $\wp(z, L)$ at $z = 0$ is

$$\wp(z, L) = \frac{1}{z^2} + \sum_{n=1}^{\infty} (2n+1) G_{2n+2}(L) z^{2n}$$

3 Lattices define elliptic curves

In this section we will show how to link a lattice to an elliptic curve.

Theorem 3.1. Let L be a lattice. The function $\wp(z, L)$ satisfies the differential equation

$$\wp'(z, L) = 4\wp(z, L)^3 - g_2(L)\wp(z, L) - g_3(L)$$

where $g_2(L) = 60G_4(L)$ and $g_3(L) = 140G_6(L)$.

Proof. We may apply theorem 2.8 to compute the first few terms of the Laurent series expansions for $\wp(z)$ and $\wp'(z)$ at $z_0 = 0$:

$$\begin{aligned}\wp(z) &= \frac{1}{z^2} + 3G_4(L)z^2 + 5G_6(L)z^4 + \dots \\ \wp'(z) &= -\frac{2}{z^3} + 6G_4(L)z + 20G_6(L)z^3 + \dots \\ \wp(z)^3 &= \frac{1}{z^6} + \frac{9G_4(L)}{z^2} + 15G_6(L) + \dots \\ \wp'(z)^2 &= \frac{4}{z^6} - \frac{24G_4(L)}{z^2} - 80G_6(L) + \dots\end{aligned}$$

Now let

$$f(z) = \wp'(z)^2 - 4\wp(z)^3 + 60G_4(L)\wp(z) + 140G_6(L).$$

We can compute the Laurent series expansion for $f(z)$ at $z_0 = 0$ as a linear combination of those computed above, and one finds that the non-positive powers of z all cancel; we thus have $f(0) = 0$.

Moreover we have that $f(z)$ is holomorphic on the fundamental parallelogram F_0 , indeed $\wp$ and $\wp'$ have pole only in L and $F_0 \cap L = \{0\}$, then we have only to check that $f(z)$ is holomorphic in 0. In the previous lines we have already seen that the laurent expansion of f in 0 has no negative powers of z , then it is holomorphic in 0.

Since both $\wp$ and $\wp'$ are periodic, also f is periodic, then we can cover the complex plane with fundamental parallelograms such that f is holomorphic in each parallelogram. Thus f is holomorphic in $\mathbb{C}$.

Note also that $f(z)$ is bounded since all values attained by f are attained on the closure of a fundamental parallelogram, which is a compact set². It then follows from Liouville's Theorem that f is a constant function and hence identically zero (since we have proved that $f(0) = 0$). $\square$

With $y = \wp'(z, L)$ and $x = \wp(z, L)$, from the differential equation we obtain the curve

$$y^2 = 4x^3 - g_2(L)x - g_3(L) \tag{1}$$

Let's check for which point the curve (1) is singular: let us take the projective curve obtained by (1)

$$zy^2 = 4x^3 - g_2(L)xz^2 - g_3(L)z^3$$

²Since f is periodic, we have that on the boundary of a fundamental parallelogram A that is not in A , f is defined in another parallelogram.

and let's compute all the partial derivatives

$$2zy = 0, \quad 12x^2 - g_2(L)z^2 = 0 \quad \text{and} \quad y^2 + 2g_2(L)xz + 3g_3(L)z^2 = 0$$

If $z = 0$, then also $x = y = 0$ that is not possible. Then, let us assume $z = 1$, and then from the first equation above we obtain $y = 0$. The third equation forces $x = -\frac{3}{2g_2(L)}$. Plugging x and z in the first equation we obtain $27g_3(L)^2 - g_2(L)^3 = 0$, thus so long as

$$\Delta(L) := g_2(L)^3 - 27g_3(L)^2$$

is non-zero, (1) defines a non-singular curve on $\mathbb{C}$, thus an elliptic curve. Luckily we have that the curve (1) is an elliptic curve for each lattice L , indeed one can prove that:

Lemma 3.2. For any L lattice, we have that $\Delta(L) \neq 0$.

Finally we can give the following definition:

Definition 3.3. Given L a lattice the following equation

$$E_L : y^2 = 4x^3 - g_2(L)x - g_3(L)$$

describes a non-singular elliptic curve. Moreover the following map

$$\begin{aligned} \phi_L : \mathbb{C}/L &\rightarrow E_L \\ [z] &\mapsto \begin{cases} (\wp(z, L), \wp'(z, L)) & , \text{ if } z \notin L \\ \mathcal{O} & , \text{ if } z \in L \end{cases} \end{aligned}$$

maps point from the complex torus on the elliptic curve ³.

One can also prove that the previous map is an isomorphism of abelian group, in the following lines we will only prove that is bijective and that preserves the inverses and the identity. To do this we need two lemmas:

Lemma 3.4. $\Phi|_{\mathbb{C}/L[2]}$ is an isomorphism between $(\mathbb{C}/L)[2]$ and $E[2]$.

Lemma 3.5. Given an elliptic function f on a lattice L , we have that the number of zeros counted with multiplicity in $\mathcal{F}_\alpha$ (a fundamental parallelogram) is equal to number of poles of f in $\mathcal{F}_\alpha$.

Lemma 3.6. A point $z \notin L$ is a zero for $\wp'(z, L)$ if and only if $2z \in L$.

Proof. ($\Leftarrow$) Let $2z \in L$ for some $z \notin L$. Since $\wp'$ is both periodic to respect L and also odd, we have that

$$\wp'(z) = \wp'(z - 2z) = \wp'(-z) = -\wp'(z)$$

³One can also prove that ϕ_L is both an isomorphism of additive group and an isomorphism of complex manifolds, hence an isomorphism of Lie groups.

then $\wp'(z) = 0$.

($\Rightarrow$) Assume now that $\wp'(z, L) = 0$ and let $L = [w_1, w_2]$ for some $w_1, w_2 \in \mathbb{C}$. Then the only points in $\mathcal{F}_0$ that are not in L and that satisfy $2z \in L$ are

$$\frac{w_1}{2}, \frac{w_2}{2}, \frac{w_1 + w_2}{2}$$

Since $\wp'$ has 3 poles in $\mathcal{F}_0$, by lemma 3.5 we have that it has only 3 zeros and they are the points that we have already found. $\square$

Now we can prove the theorem:

Theorem 3.7. Φ is bijective and preserves the inverses and the identity.

Proof. It is clear that $\Phi(0) = \mathcal{O}$, so the identity is preserved.

Let $z \notin L$, since $\wp(z, L)$ is odd and $\wp'(z, L)$ is even, we have that

$$\Phi(-z) = (\wp(-z, L), \wp'(-z, L)) = (\wp(z, L), -\wp'(z, L)) = -\Phi(z)$$

then Φ preserves also the inverses.

Now let's prove that Φ is injective, then let $z_1, z_2 \in \mathcal{F}_0$ such that $\Phi([z_1]) = \Phi([z_2])$. By lemma 3.4 we have that Φ is injective on the point of order 2, then we can assume that $2z_1 \notin L$. Now by lemma 3.6 we have that $\wp'(z_1, L) \neq 0$. We can now define the map $f(z) := \wp(z, L) - \wp(z_1, L)$, this map has 2 poles in $\mathcal{F}_0$, then, using lemma 3.5 we have that it has only 2 zeros in $\mathcal{F}_0$. One of the zeros is clearly z_1 , moreover, since $\wp$ is even, we have that the other zero is $-z_1$. Now since $\Phi([z_1]) = \Phi([z_2])$, we have that $\wp(z_1, L) = \wp(z_2, L)$ and then we also have that z_2 is a zero of f . Since we said that the zeros of f in $\mathcal{F}_0$ are 2, we have that $z_2 \equiv \pm z_1 \pmod{L}$. Moreover we have also that $\wp'(z_1, L) = \wp'(z_2, L)$, since $\wp'$ is odd, we have that

$$\wp'(-z_1, L) = -\wp'(z_1, L) \neq \wp'(z_1, L) = \wp'(z_2, L)$$

that forces $z_2 \equiv z_1 \pmod{L}$.

Now we want to prove the surjectivity. Let $(x_0, y_0) \in E(\mathbb{C})$. As we have done above, we can define $f(z) := \wp(z, L) - x_0$, and using again lemma 3.5 we have that there are only two zeros for f in $\mathcal{F}_0$. Moreover, f has got a pole in 0, then a zero $z_0 \in \mathcal{F}_0$ for f is such that $z_0 \neq 0$. Now we have that

$$\Phi([z_0]) = ((\wp(z_0), \wp'(z_0))) = (x_0, \wp'(z_0))$$

since (x_0, y_0) is a point of the elliptic curve, we have that $\wp'(z_0) = \pm y_0$. If $\wp'(z_0) = y_0$, we have done. Moreover, if $\wp'(z_0) = -y_0$ we have that

$$\Phi(-[z_0]) = -\Phi([z_0]) = -(x_0, \wp'(z_0)) = -(x_0, -y_0) = (x_0, y_0)$$

Then the map is bijective. $\square$

One can also prove the vice-versa, i.e. given $E/\mathbb{C}$ an elliptic curve we can find a lattice L such that E_L is isomorphic to L . This theorem is not easy to prove and requires more tools as the *homotheties* and *modular groups*.

⁴I need to use the congruence since $-z_1$ does not lie in $\mathcal{F}_0$.